

INTRODUCTORY PAPER



2017 SCIENCE

**DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.**

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

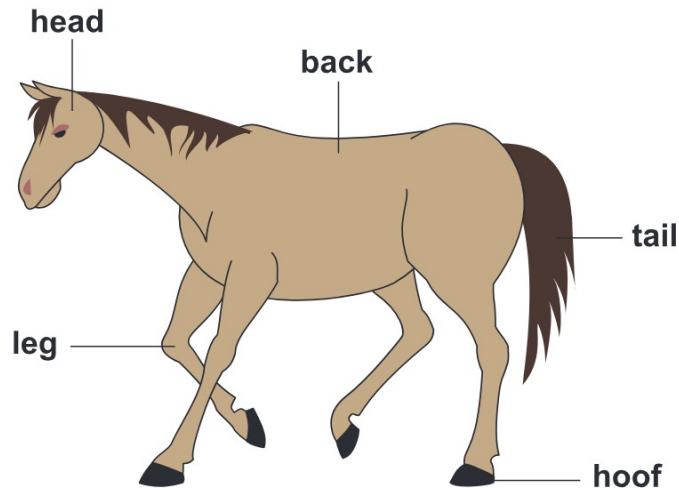
Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

You may use a calculator and a ruler.

1. The drawing shows a horse.



What is the hoof joined to?

- (A) tail (B) back (C) leg (D) head

For questions 2 and 3 use the information below.

Jackie recorded the weather on the chart for one school week.

Monday	Tuesday	Wednesday	Thursday	Friday
				

KEY			
			
Sun	wind	cloud	snow

2. On how many days did Jackie see the Sun?
- (A) 1 (B) 2 (C) 3 (D) 4
3. What season is it most likely to be?
- (A) Spring
(B) Summer
(C) Autumn
(D) Winter

4. This is a goat.



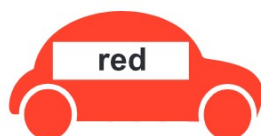
Here are four animals.



How many of the animals have the same number of legs as the goat?

- (A) 0 (B) 1 (C) 2 (D) 3
5. Wayne, Chloe, Max and Suma had four toy cars. They were all the same except for their colour.

Wayne



Chloe



Max



Suma



They each gave their car one push at the start of the same track. Wayne's car travelled the furthest.

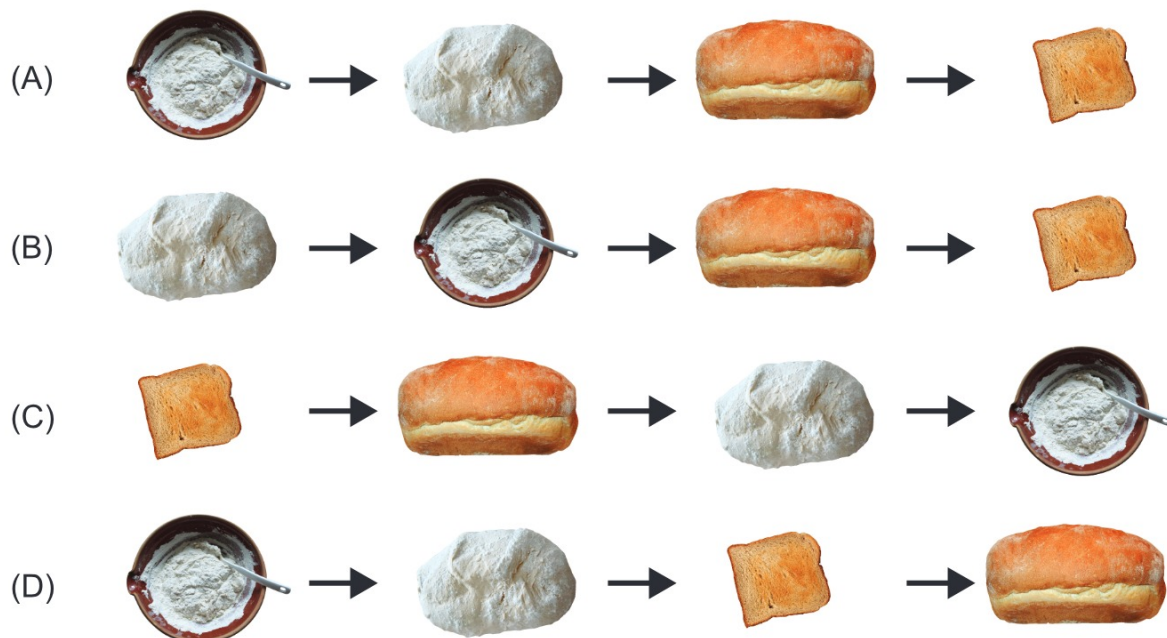
Why did Wayne's car travel the furthest?

- (A) Wayne's car was red.
(B) Wayne's car was bigger.
(C) Wayne's track was longer.
(D) Wayne pushed the hardest.

6. Angela's teacher gave the class a list of steps for making a slice of toast.

- mix flour, salt and yeast
- add water and mix to make dough
- bake the dough in the oven until golden brown
- cut off a slice and toast it

Which of the following shows the correct order for making a slice of toast?



7. Everything around us is made up of materials. The different properties of materials make them useful for different things.

The table lists some properties of four materials.

Material	Properties
leather	soft, absorbs water
wood	hard, strong
rubber	soft, waterproof
fabric	soft, lets water through

What material would be best to make dish-washing gloves?

- (A) leather
(B) wood
(C) rubber
(D) fabric

8. Landforms are the natural features that exist on Earth's surface. The diagrams show some examples of landforms.



mountains



island



jungle



ocean



desert



river



volcano



waterfall

Nilesh lives on an area of land surrounded by water.

Which of these describes where Nilesh lives?

- (A) in the mountains
 - (B) on an island
 - (C) in the desert
 - (D) near a river
9. Ruisi works at a panda sanctuary. One of her duties is to check that the newborn pandas are growing at a normal rate.

The table lists how much pandas should weigh at different ages.

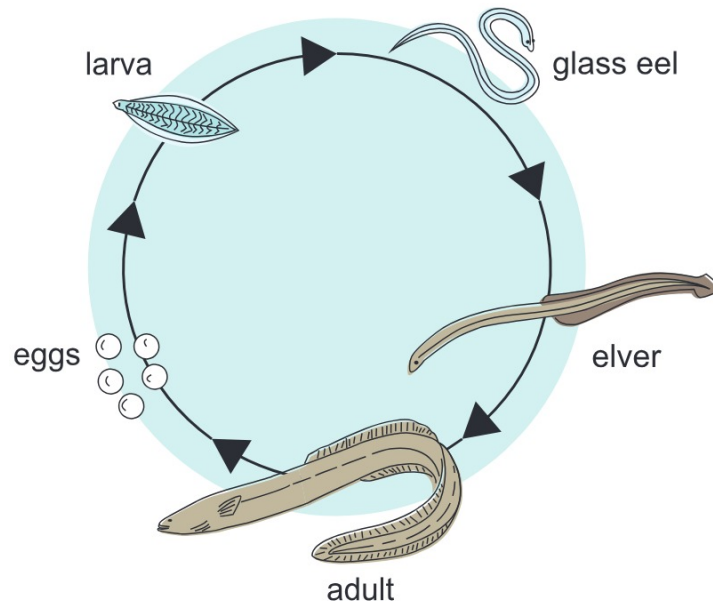
Age	Expected mass (kilograms)
1 week	1
4 weeks	2
12 weeks	5
52 weeks	40

Ruisi is weighing a panda cub that is 6 weeks old.

How much is the panda cub expected to weigh?

- (A) less than 1 kilogram
- (B) between 1 and 2 kilograms
- (C) between 2 and 5 kilograms
- (D) more than 5 kilograms

10. The diagram shows the life cycle of an eel.

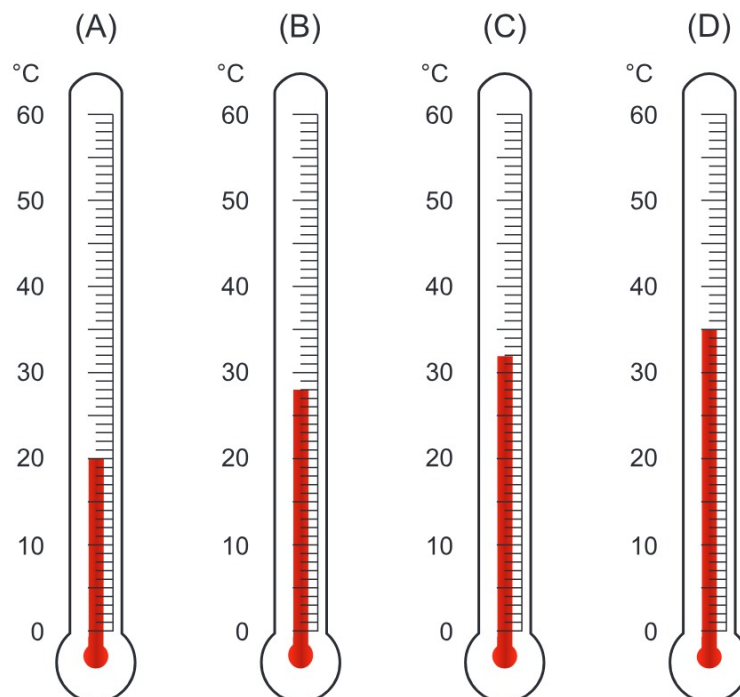


What is an eel called when it hatches out of an egg?

- (A) adult
- (B) elver
- (C) glass eel
- (D) larva

11. Adam has four thermometers.

Which thermometer shows a temperature of 32 °C?



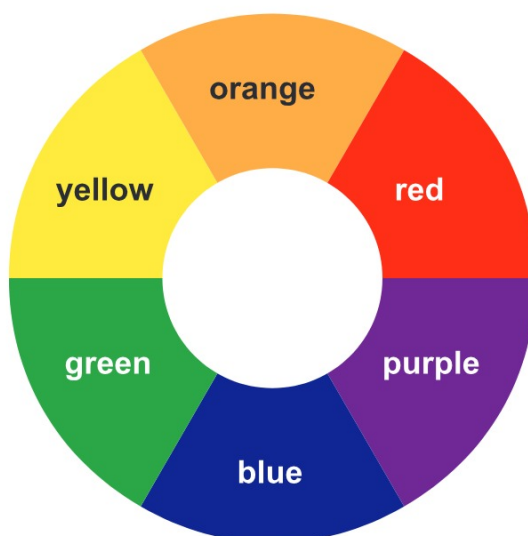
2. Lisa found a frog in her garden. She held her finger near the frog and took this photo.



What does her finger help to show?

- (A) the size of the frog
(B) the colour of the frog
(C) the age of the frog
(D) the shape of the frog
13. Red, blue and yellow are three colours that cannot be made by mixing other colours together. They are called the primary colours. Mixing two primary colours together makes a secondary colour.

In the colour wheel, each secondary colour is made by mixing the two primary colours on either side of it.



Which two primary colours make purple when you mix them?

- (A) orange and green
(B) yellow and white
(C) yellow and red
(D) red and blue

14. Joe has a tree in his paddock.



What causes the tree to have a shadow?

- (A) The shadow is part of the tree.
- (B) The top of the tree is close to the sun.
- (C) The tree needs a shadow to keep it cool in the sun.
- (D) The tree stops the sunlight from reaching the ground.

15. Priya found some broccoli at the bottom of the fridge and wanted to know if it is still safe to eat.



Priya's mum told her that when broccoli goes bad it:

- produces a strong odour
- changes colour from bright green to yellow
- loses its crisp texture and turns soft.

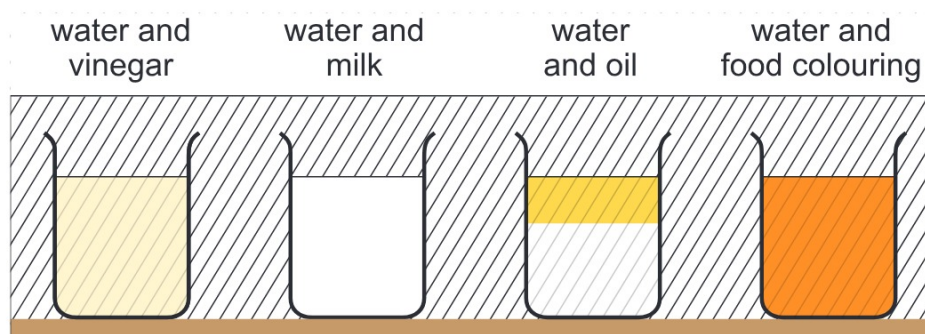
Which senses does Priya need to use to determine if her broccoli is still safe to eat?

- (A) sight, smell, touch
- (B) smell, taste, touch
- (C) hearing, sight, taste
- (D) smell, touch, hearing

For questions 16 and 17 use the information below.

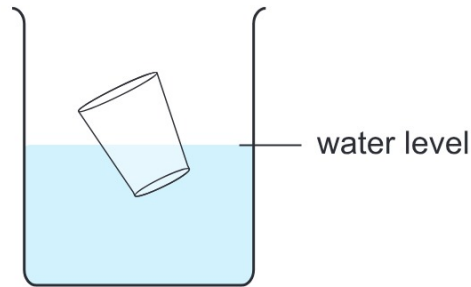
A solution is made when two liquids mix and become a clear liquid. An emulsion is made when two liquids mix but become a cloudy liquid. Some liquids do not mix but form separate layers.

Jack mixed some liquids with water and placed them in front of a patterned board.



16. Which of these makes an emulsion when mixed with water?
- (A) vinegar
 - (B) milk
 - (C) oil
 - (D) food colouring
17. What do the vinegar, oil and food colouring have in common when mixed with water?
- (A) They have no taste.
 - (B) They are clear liquids.
 - (C) They can be used as plant food.
 - (D) They can be used as colours for drawing.
18. Ben lives in South Africa. Ben's dad tells him that from January the daylight hours will get less every day until midyear.
- What does Ben need to do every day to find out if this is true?
- (A) Record the time he goes to bed and the time he wakes up.
 - (B) Record the time when the Sun is right overhead.
 - (C) Record the morning and evening temperatures.
 - (D) Record the time of sunrise and sunset.

19. Lisa placed an empty plastic cup in a jar of water. The plastic cup floated in the water.



What would happen to the water level in the jar if Lisa placed a marble in the plastic cup?

- (A) The water level would be lower. (B) There would be more water in the jar.
(C) The water level would move upwards. (D) The water level would not change.

20. This is a photograph of the sole of one of Bernie's shoes.



Which of these is another photograph of the sole of the same shoe?

(A)



(B)



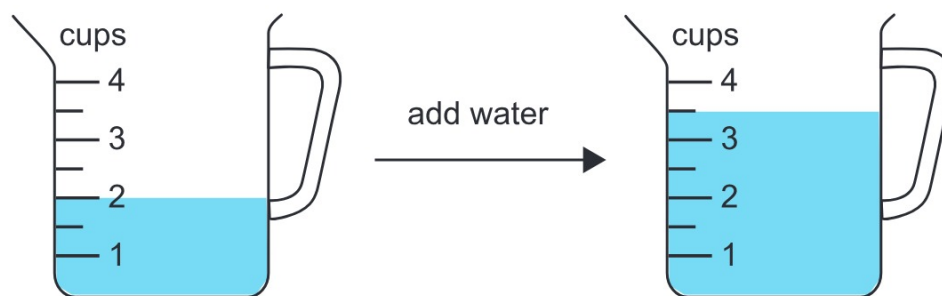
(C)



(D)



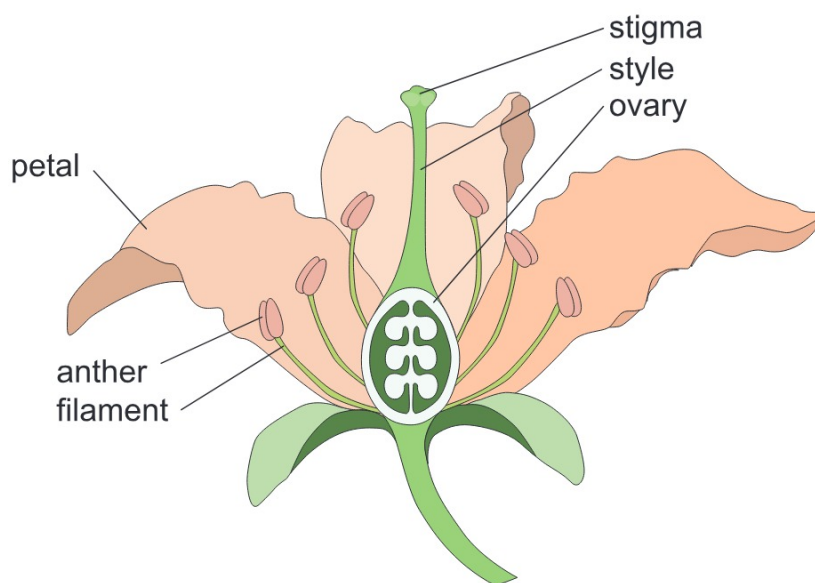
21. Lu had a jug containing some water. She then added some more water to it.



How many cups of water did Lu add?

- (A) 1
- (B) $1\frac{1}{2}$
- (C) 2
- (D) $3\frac{1}{2}$

22. This diagram shows some parts of a flower.



On top of which part of the flower is the anther found?

- (A) stigma
- (B) filament
- (C) style
- (D) ovary

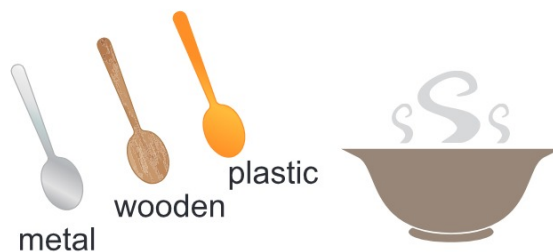
For questions 23 and 24 use the information below.

The diagrams show some of the steps in making cotton fabric from cotton plants.



23. According to the diagram, how many steps does it take to turn cotton bolls into a cotton shirt?
- (A) 2 (B) 3 (C) 4 (D) 5
24. What happens during the step of spinning?
- (A) Cotton bolls are turned into cotton fibres.
(B) Cotton fibres are turned into yarn.
(C) Yarn is turned into fabric.
(D) Fabric is turned into clothes.

25. Allan had three spoons: a metal spoon, a wooden spoon and a plastic spoon.
He placed the spoons into a bowl of hot water.



What question can Allan answer from this test?

- (A) Which spoon heats up the fastest?
(B) Which spoon cools down the water the most?
(C) Does hot water clean spoons better than cold water?
(D) How many spoons can he fit in the bowl?

26. Wendy blew up a balloon in her home. Her home was very cold.

She then took the balloon outside and left it in the hot sunlight for a while. She noticed that it became bigger.

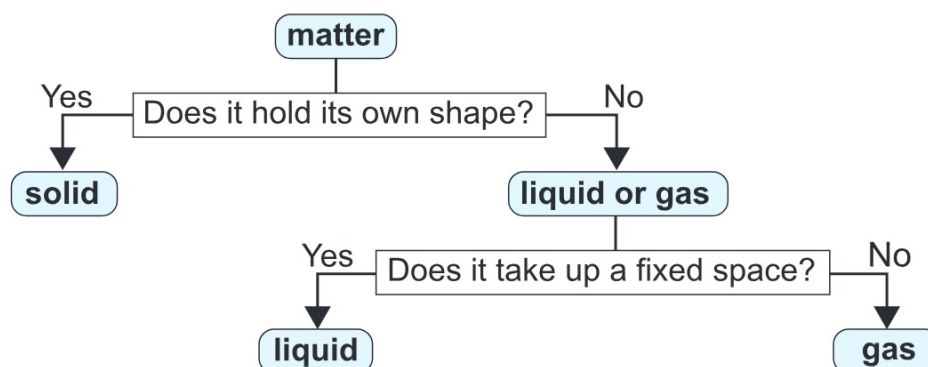


What is the best explanation for this?

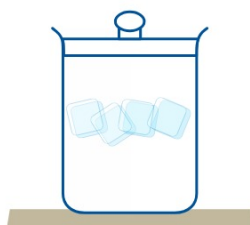
- (A) The balloon stretches more when it gets warmer.
- (B) The sunlight makes some air leak into the balloon.
- (C) The colour of the balloon makes it appear to be bigger in the sunlight.
- (D) The air inside the balloon takes up more space when it gets warmer.

27. Everything around us is made up of matter. Matter can exist in three main states: solid, liquid or gas. For example, wood is a solid, water is a liquid, and air is a gas.

The flowchart helps to determine whether something is a solid, a liquid or a gas.



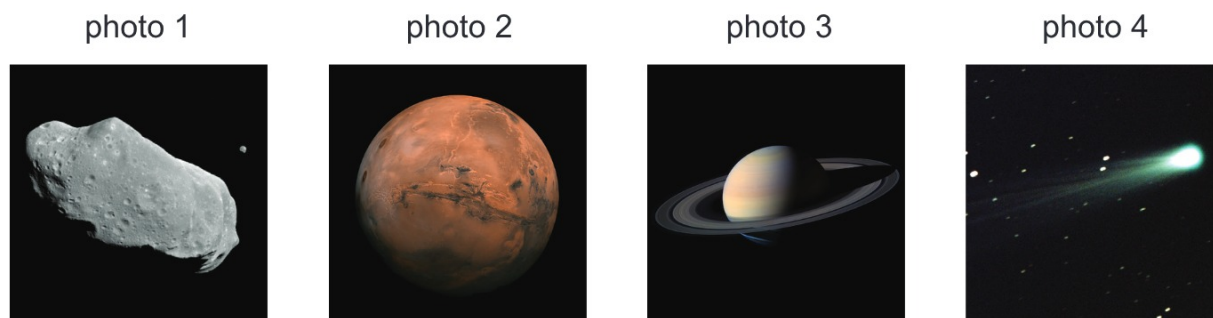
Karin has some ice cubes in a cup of drink.



What states of matter are inside this cup?

- | | |
|-------------------------|---------------------------|
| (A) liquid and gas only | (B) liquid and solid only |
| (C) solid and gas only | (D) solid, liquid and gas |

28. Paula's teacher gave her four photos, each one showing a different object in space.



She also told Paula four facts. Each fact was about one of these objects in space.

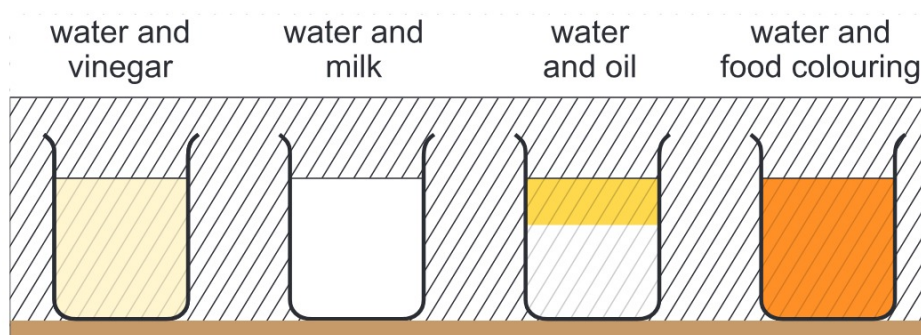
- Saturn is a planet with rings around it.
- A comet is a ball of frozen gases, leaving a trail behind it.
- A meteoroid is a lump of rock travelling in space.
- Mars is a red planet.

Which photo shows a meteoroid?

- (A) photo 1 (B) photo 2 (C) photo 3 (D) photo 4

29. A solution is made when two liquids mix and become a clear liquid. An emulsion is made when two liquids mix but become a cloudy liquid. Some liquids do not mix and form separate layers.

Jack mixed some liquids with water and placed them in front of a patterned board.

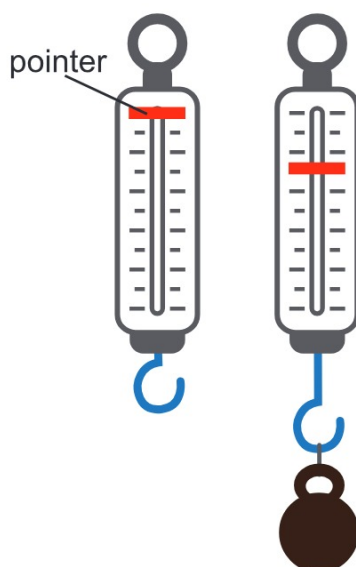


Which of these makes an emulsion when mixed with water?

- (A) vinegar
(B) milk
(C) oil
(D) food colouring

30. A spring balance can be used to weigh things. A spring inside the balance stretches when an object hangs from it. The pointer that attaches to the spring points to how much the object weighs.

This diagram shows what happens when an object hangs from a spring balance.



Which diagram correctly shows the balance with two of these objects hanging from it?

